USN | | | | | | | | | | [23TBSMA21]

SECOND SEMESTER B.Tech EXAMINATION, APRIL - 2024

PROBABILITY AND VECTOR SPACES

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART - A			**05 x02=10**
1.	**a.**	Define exhaustive events with a suitable example.	L2	CO1	**02**
	b.	If X is a random variable with the range X = 1,2,3,4,5, and f(X) = (X −2)/5. Determine whether f(X) can serve as a probability distribution.	L1	CO2	**02**
	c.	Define subspace.	L1	CO3	**02**
	d.	Find the matrix of linear transformation $T: V_3(R) \to V_2(R)$ defined by $T(x,y,z) = (x+y,\ y+z)$.	L2	CO4	**02**
	e.	Express matrix A into singular value decomposition.	L1	CO6	**02**
		PART – B			**06X05=30**
2.	**a.**	A die is rolled twice, Find the probability that the sum of the numbers in the two faces of the dice is **i.** an odd number **ii.** an even number **iii.** greater than 10.	L3	CO1	**05**
		OR			
	b.	There are 10 students of which three are graduates, if a committee of five is to be formed, Find the probability that there are **i.** only 2 graduates **ii.** at least 2 graduates.	L3	CO1	**05**
3.	**a.**	20 wristwatches in a box of 100 are defective. If 10 watches are selected at random. Find the probability that **i.** 10 are defective. **ii.** 10 are good. **iii.** at least one watch is defective.	L3	CO2	**05**
		OR			
	b.	If X is Uniform between -2 and 6 then find the **i.** mean of X. **ii.** variance of X. **iii.** probability that X takes a value less than 4.	L3	CO2	**05**
4.	**a.**	Show that the vector $(3,-7,6)$ is linear combination of vectors in S, Where $S = \{(1,-3,2), (2,4,1), (1,1,1)\}$ belongs to $V_3(R)$	L3	CO3	**05**
		OR			
	b.	Show that the subset $S = (x,\ 2x,\ -3x,\ x)$is a subspace of $V_4(R)$.	L3	CO3	**05**
5.	**a.**	Examine whether the transformation $T: R^2 \to R^3$ defined by $T(x,y) = (x-y,\ y,\ x+y)$ is a linear transformation or not?	L3	CO4	**05**
		OR			

	b.	Use Python program to check whether the given vectors $v_1 = \begin{bmatrix} 2 \\ -4 \\ 8 \end{bmatrix} \& v_2 = \begin{bmatrix} 4 \\ 10 \\ 4 \end{bmatrix}$ are orthogonal or not?	L3	CO4	**05**
6.	**a.**	Determine the Eigenvalues of the matrix $A = \begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}$.	L3	CO6	**05**
		OR			
	b.	The data between the number of hours studied by a student and the scores achieved on a test is given below. Find the equation of the straight line by using least square method.	L3	CO5	**05**

Hours Studied	2	3	4	5	6	7
Test Score	65	70	75	80	85	90

7.	**a.**	Use Python code to find the second-degree polynomial $y = p(x) = a_0 + a_1x + a_2x^2$ that includes the points (1, 5), (2, 13), and (3, 25).	L3	CO5	**05**
		OR			
	b.	Use the induced matrix associated with each transformation to find the new point. **i.** $x = (5,0)$ Rotated 90° in the counter-clockwise direction. **ii.** $x = (0,5,1)$ Rotated 90° in the counter-clockwise direction about the y-axis.	L3	CO4	**05**
		PART - C			**06X10=60**
8.	**a.**	i. State Bayes Theorem. ii. Three machines A, B and C produce respectively 60%, 30%, 10%, of the total number of items of a factory. The percentages of defective output of these machines are respectively 2%, 3% and 4%. An item is selected at random and is found defective. Find the probability that the item was produced by machine C.	L3	CO2	**10**
		OR			
	b.	Given $P(A)=3/4$, $P(B)=1/5$ *and* $P(A \cap B)=1/20$, Find $P(\bar{A} \cap B)$, $P(B\|A)$, $P(\bar{A}\|\bar{B})$, $P(\bar{B}\|\bar{A})$, *and* $P(\bar{A}\|B)$.	L3	CO1	**10**
9.	**a.**	A random variable X has the following probability function for various values of X.	L3	CO2	**10**

X	-3	-2	-1	0	1	2	3
$p(X)$	k	$2k$	$3k$	$4k$	$3k$	$2k$	k

i. Find k .
ii. Mean, Variance and Standard Deviation.
iii. Evaluate $p(x \leq 1)$, $p(x > 1)$ *and* $p(-1 < x \leq 2)$.

OR

	b.	In a normal distribution 31% of the items are under 45 and 8% are over 64. Find the mean and standard deviation of the distribution. Given $\emptyset(0.5) = 0.19$ and $\emptyset(1.4) = 0.42$.	L3	CO2	**10**
10.	**a.**	Show that the set V is a vector space over the field F of real numbers mentioned under usual addition and scalar multiplication where $V = \left\{\begin{pmatrix} x & 0 \\ 0 & y \end{pmatrix}, x, y \in R\right\}$.	L3	CO3	**10**
		OR			
	b.	Determine the four fundamental subspaces for the matrix A, where $A = \begin{bmatrix} 1 & 3 & 2 \\ 2 & 6 & 4 \\ 0 & 1 & 1 \end{bmatrix}$	L3	CO3	**10**
11.	**a.**	Let W be the subspace of R^4 with basis $S = \{u_1, u_2, u_3\}$ where $u_1 = (1, -2, 0, 1)$, $u_2 = (-1, 0, 0, -1)$ and $u_3 = (1, 1, 0, 0)$. Use Gram-Schmidt process to transform S to orthonormal basis for W.	L3	CO5	**10**
		OR			
	b.	If $T: R^2 \to R^2$ is a linear transformation such that $T(1, 0) = (1, 1)$ *and* $T(0, 1) = (-1, 2)$. Show that T maps the square with vertices $(0, 0)$, $(1, 0)$, $(1, 1)$ *and* $(0, 1)$ into a parallelogram.	L3	CO4	**10**
12.	**a.**	The following data shows atmospheric pollutants y_i (relative to an EPA standard) at half hour intervals t_i. Find the quadratic equation to fit the data.	L3	CO5	**10**

t_i	1	1.5	2	2.5	3	3.5	4	4.5	5
y_i	-0.15	0.24	0.68	1.04	1.21	1.15	0.86	0.41	-0.08

		OR			
	b.	Use Python code to reduce the following matrix A into diagonal form. $A = \begin{bmatrix} 3 & -2 & 4 & -2 \\ 5 & 3 & -3 & -2 \\ 5 & -2 & 2 & -2 \\ 5 & -2 & -3 & 3 \end{bmatrix}$	L3	CO6	**10**
13.	**a.**	Write Python code with a suitable example for **i.** Union and Intersection operations over sets. **ii.** Addition theorem of probability for two events.	L3	CO1	**10**
		OR			
	b.	Determine Singular value decomposition of the matrix $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$.	L3	CO6	**10**